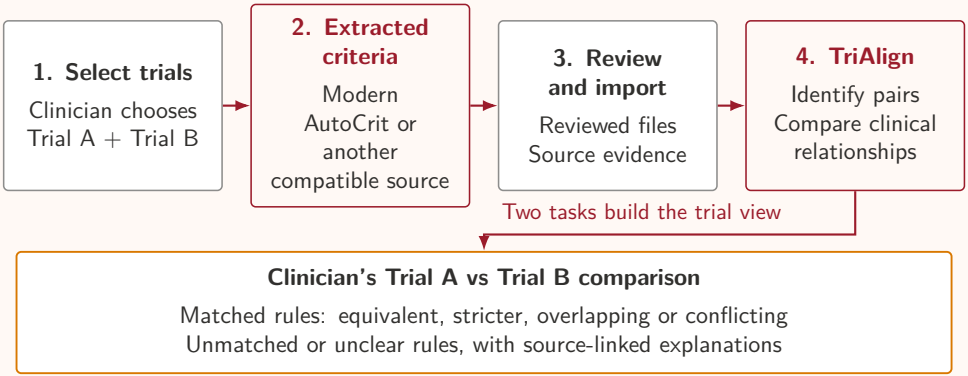
TriAlign

Progress Overview & Blockers

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Current workflow: compare two trials



Now working: load the frozen corpus or import extracted criteria, compare two trials, inspect evidence and save a review.

TriAlign runs independently of Modern AutoCrit. Direct extraction inside TriAlign remains future work.

Data progress: the reviewed corpus is frozen

Artifact	Current state
Criterion corpus	**3,537 criteria across 100 trials**, replacing the proposal's preliminary extraction count.
Reviewed source	`trial100_v1.xlsx`; criterion-level review completed by the project owner.
Frozen snapshot	Parquet plus a checksum manifest for reproducible inputs.
Preserved evidence	Source text, parent statement, trial ID, clinical fields and context.

Important distinction: the extraction review is complete. Human validation of criterion-pair labels remains in progress.

Dataset Visualization

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	source	parent_statement	criterion_type	domain	entity	comparator	value	unit	value_kind	lower_bound	lower_inclusive	upper_bound	upper_inclusive	negated	temporal	repetition	qualifier	logical_operator	review_status
2	Aged 18 - 80 years old	Aged 18 - 80 years old	inclusion	person	Age		18 to 80	years	interval	18	☑	80	☑	☐				standalone	corrected
3	Able to give informed c	Able to give informed consent	inclusion	person	Informed Consent		present		categorical					☐				standalone	corrected
4	Available for the whole	Available for the whole durat	inclusion	observation	availability for study duration		present		categorical					☐				standalone	corrected
5	Life expectancy \>2 yea	Life expectancy \>2 years, in t	inclusion	observation	Life Expectancy	>	2	years	numerical	2	☐			☐				standalone	corrected
6	Current systemic thera	Current systemic therapy for t	exclusion	drug	current systemic therapy for bladc		present		categorical					☐				standalone	corrected
7	History of malignancy o	History of malignancy of othe	exclusion	condition	malignancy of other organ system		present		categorical					☐	History of; within past 5 years; at least 24 months			or	corrected
8	History of malignancy o	History of malignancy of othe	inclusion	measurement	time since nephroureterecto	>=	24	months	numerical	24	☑			☐	after nephroureterectomy		exception to other-organ r	or	corrected
9	Eastern Cooperative On	Eastern Cooperative Oncolog	inclusion	observation	ECOG Performance Status		0 to 3		interval	0	☑	3	☑	☐				standalone	corrected
10	Serum creatinine ≤ 2.0 r	Serum creatinine ≤ 2.0 mg/dl.	inclusion	measurement	Serum Creatinine	<=	2	mg/dl	numerical			2	☑	☐				standalone	corrected
11	Subjects with detectabl	Subjects with detectable level	exclusion	condition	detectable levels of endogenous ar		present		categorical					☐			at the time of screening	standalone	corrected
12	In the absence of rapidl	In the absence of rapidly prog	inclusion	measurement	time since prior cytotoxic tr	>=	2	weeks	numerical	2	☑			☐	before study drug administration		in the absence of rapidly p	or	corrected
13	Histologically confirme	Histologically confirmed new	inclusion	condition	GBM		present		categorical					☐				standalone	corrected
14	Supratentorial tumor lc	Supratentorial tumor locatio	inclusion	condition	supratentorial tumor location		present		categorical					☐				standalone	corrected
15	Planned treatment with	Planned treatment with adju	inclusion	drug	TMZ		150 to 200	mg/m^2 dail	interval	150	☑	200	☑	☐				standalone	corrected
16	All patients with sufficie	All patients with sufficient tis	inclusion	observation	mgmt promoter methylation dete		present		categorical					☐	prior to enrollment			standalone	corrected
17	Karnofsky Performance	Karnofsky Performance Status	inclusion	measurement	Karnofsky Performance Stat	>=	70		numerical	70	☑			☐				standalone	corrected
18	Are surgically sterile, OI	Are surgically sterile, OR	inclusion	person	surgically sterile		present		categorical					☐				or	corrected
19	Unstable angina at regis	Unstable angina at registratio	exclusion	condition	unstable angina		present		categorical					☐	at registration			standalone	corrected
20	Transmural myocardial	Transmural myocardial infarc	exclusion	condition	transmural myocardial infarction		present		categorical					☐	within the last six months prior to registration			standalone	corrected
21	Patents treated on any (	Patents treated on any other t	exclusion	person	patients treated on any other ther		present		categorical					☐	within 30 days prior to registration			standalone	corrected
22	Inability to undergo MF	Inability to undergo MRI.	exclusion	device	inability to undergo mri		present		categorical					☐				standalone	corrected
23	Are postmenopausal for	Are postmenopausal for at lea	inclusion	measurement	duration of postmenopausal	>=	1	year	numerical	1	☑			☐	before the screening visit		deterministically parsed v	or	corrected
24	Previously treated with	Previously treated with brent	exclusion	drug	brentuximab vedotin		present		categorical					☐				or	corrected
25	Previously treated with	Previously treated with brent	exclusion	drug	immune-oncology agents		present		categorical					☐				or	corrected
26	Previously treated with	Previously treated with brent	exclusion	procedure	allogeneic stem cell transplant		present		categorical					☐				or	corrected
27	Previously treated with	Previously treated with brent	exclusion	procedure	autologous stem cell transplant		present		categorical					☐				or	corrected
28	Documented history of a	Documented history of a care	exclusion	observation	cerebral vascular event		present		categorical					☐	history of			standalone	corrected
29	History of progressive n	History of progressive multifo	exclusion	condition	progressive multifocal leukoence		present		categorical					☐	History of			standalone	corrected
30	Hemoglobin ≥ 8.0 g/dl	Hemoglobin ≥ 8.0 g/dl	inclusion	measurement	Hemoglobin	>=	8	g/dl	numerical	8	☑			☐				standalone	corrected
31	Absolute neutrophil co	Absolute neutrophil count ≥ 1	inclusion	measurement	Absolute Neutrophil Count	>=	1000	mcL	numerical	1000	☑			☐				standalone	corrected
32	Platelet count ≥ 75,000	Platelet count ≥ 75,000/mcL	inclusion	measurement	Platelet Count	>=	75000	mcL	numerical	75000	☑			☐				standalone	corrected
33	Aspartate transaminase	Aspartate transaminase (AST)	inclusion	measurement	Aspartate Aminotransferase	<=	2.5		numerical			2.5	☑	☐				standalone	corrected
34	Alanine transaminase (A	Alanine transaminase (ALT) (SC	inclusion	measurement	Alanine Aminotransferase (A	<=	2.5		numerical			2.5	☑	☐				standalone	corrected
35	International normalize	International normalized ratio	inclusion	measurement	INR	<=	1.5		numerical			1.5	☑	☐				standalone	corrected
36	HBsAg negative, HBcAb,	HBsAg negative, HBcAb negat	inclusion	condition	hbsab positive		present		categorical					☐			applies to patients with HI	standalone	corrected

Each row retains the criterion text and parent statement alongside its role, clinical concept, numeric bounds, timing, qualifiers and review status.

How we keep related trials out of the test set

Term	Meaning in our dataset
Trial family	Related registrations, such as a main study and its extension. Keep related records together to avoid learning from a near-copy of a test case.
Partition	A dataset split: **68 trials for training**, **15 for validation** and **15 for testing**. Families stay in one split.
Quarantine	**Two trials** have unresolved parent-study links. Exclude them from training, validation and testing until those links are resolved.
Exposure log	Records cases already inspected during development or demos. The reservation list keeps 39 trials in training and two in quarantine, protecting held-out evaluation.

Pair review: 75 initial pairs plus 300 new candidates = 375 pairs. Of these, 370 are eligible training pairs and five historical pairs are excluded.

Validation/test *trials* are assigned; their pair labels still need review.

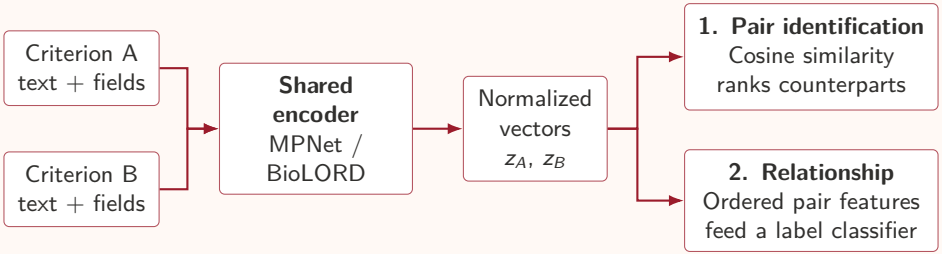
Why these three retrieval baselines?

Method	Representation	Why we chose it
TF–IDF	**Lexical** word weights	A word-overlap reference: does shared wording already find useful counterparts?
MPNet	**General** embeddings	A general sentence-similarity model. Tests matching beyond exact words and provides our initial encoder backbone.
BioLORD	**Biomedical** embeddings	Builds on MPNet using biomedical definitions and ontology knowledge. Tests the value of clinical terminology and concept training.

The two embedding models give a general-versus-biomedical comparison on a related backbone. Neither was chosen because it has proved best on our pairs.

Saved MPNet example: A permits treated, asymptomatic brain metastases; B excludes known brain metastases. High similarity finds the pair, while **B's blanket exclusion is more restrictive**.

One shared model supports both tasks

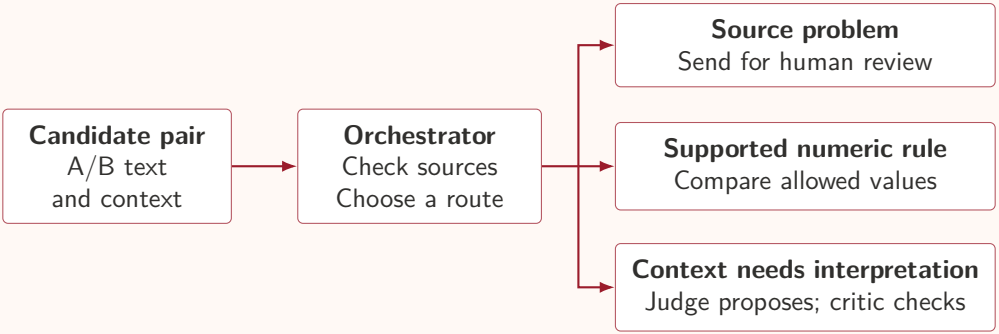


Shared encoder: the same model reads A and B, turning each criterion into a numerical description that can be compared with other criteria.

Relationship head: a small classifier compares those descriptions and their differences to propose a clinical relationship. Keeping A and B separate lets it identify which rule is more restrictive.

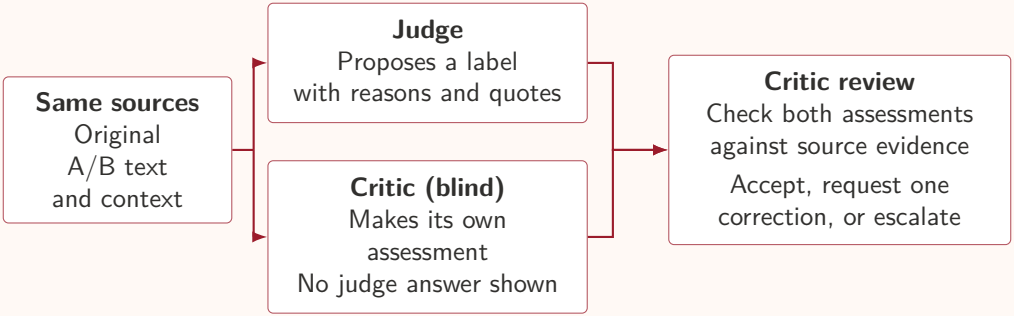
Future work: train and evaluate using human-validated pair labels.

How the orchestrator chooses a comparison route



A trained model may suggest a label for contextual review. Its suggestion still needs source evidence and critic review.

Independent judge and critic assessments



Independent first assessments: separate requests. The critic's assessment is fixed before it reviews the judge's proposal.

They may use the same model and make the same mistake. Human review remains necessary.

November 8 case: matching platelets, unresolved timing

Trial A rule summary

Platelets $\geq$ 100,000/mcL.
No assessment window stated.

Trial B rule summary

Platelets $\geq$ 100,000/mcL, measured within two weeks before Step 1 randomization.

1. The judge proposed **B more restrictive**.
2. The blind critic initially agreed with that label.
3. Final critic review rejected their shared assumption: missing timing in A does not prove that A permits a result from any time.

Final outcome: human review. Source evidence must establish a shared assessment frame before claiming one allowed population contains the other.

Limitation: a critic-accepted bilirubin explanation in the same run also used unstated timing assumptions. Accepted explanations still need adjudication.

Public benchmarks: what each would actually test

Resource	Native task	Planned comparison
NLI4CT-P SemEval 2024	Does clinical-trial evidence support or contradict a statement?	Auxiliary judge/critic test on eligibility cases, using native NLI labels and F1. An adapter is still needed.
CReSE	Recommend eligibility criteria from trial information.	Potential eligibility-specific embedding baseline on our retrieval pool, if its artifacts are reproducible.
EligMeta	Eligibility-aware trial discovery and meta-analysis.	Compare alignment and exception handling on matched cases. Requires an adapter to our task.

AMEND++ is deferred: it predicts future amendments rather than pair relationships.

These resources do not directly provide our eight-class labels. Compare published scores only when tasks, inputs and splits match.

Future work after the labeling gate

- Train the shared encoder and relationship head on approved labels; select settings using validation data.
- Run matched retrieval, relationship and agent comparisons on held-out cases.
- Tune abstention or calibration on validation data and report class errors, missed counterparts and reviewer corrections.
- Rehearse source-backed cases in the working local UI.

Longer term: direct extraction within TriAlign and comparison of whole requirement groups when extractors split the same rule differently.

These extensions remain research work, rather than demonstrated contributions.